

Scope of Work: ShopVista Data Modernization Project

1. Project Overview

ShopVista, a rapidly growing e-commerce company, faced significant challenges in managing data distributed across multiple systems and files (orders, shipments, returns, and dimension tables such as customers, products, categories, brands, and dates). This fragmented structure caused manual data reconciliation, delayed business reporting, and limited visibility into operational performance.

To address these challenges, the project aims to design and implement a centralized, automated data platform on Microsoft Azure, ensuring a single source of truth for all business reporting and analytics.

2. Objectives

- Establish a unified data architecture integrating all key datasets (orders, shipments, returns, and dimensions).
- Automate daily ingestion, transformation, and aggregation processes.
- Provide clean, reliable, and analytics-ready data using the Bronze → Silver → Gold medallion architecture.
- Enable the business team to generate real-time insights through Power BI dashboards.

3. Scope of Work

3.1 Data Architecture & Platform Setup

- Configure Azure Data Lake Storage Gen2 (ADLS) for centralized data storage.
- Set up Databricks as the data processing and transformation engine.
- Implement folder structures and access policies for Bronze, Silver, and Gold layers.

3.2 Data Ingestion (Bronze Layer)

- Automate daily ingestion pipelines to bring raw files (orders, shipments, returns) into ADLS.
- Maintain metadata for ingestion status, source file tracking, and error handling.

3.3 Data Transformation & Quality (Silver Layer)

- Define schemas and enforce data quality checks for each dataset.
- Standardize and cleanse data to ensure consistency and accuracy.
- Handle nulls, duplicates, and referential integrity across datasets.

3.4 Data Integration & Modeling (Gold Layer)

- Create integrated fact tables (Orders, Shipments, Returns) enriched with dimension data (Customers, Products, Categories, Brands, Date).
- Build a star schema suitable for analytical reporting.
- Optimize tables for Power BI consumption.

3.5 Reporting & Visualization

- Develop Power BI dashboards providing insights on:
 - Sales by Brand, Category and Country
 - Monthly Revenue Trend
 - Return Patterns by Product and Category
- Implement role-based access and refresh automation.

4. Deliverables

Category	Deliverable	Description
Data Pipeline	Automated Ingestion	Daily data loads into ADLS (Bronze layer).
Data Quality	Cleaned Datasets	Schema enforcement and validation in Silver layer.
Data Warehouse	Gold Tables	Unified fact and dimension tables ready for analytics.
Visualization	Power BI Reports	Interactive dashboards for business insights.

5. Project Outcomes

- Eliminated manual data consolidation efforts.
- Reduced reporting turnaround from hours to minutes.
- Established a single source of truth for all business reporting.
- Enabled data-driven decision-making through live dashboards.
- Built a scalable data foundation to accommodate future data growth.

6. Assumptions & Dependencies

- Source data files are consistently available in agreed formats and locations.
- Azure and Power BI licenses are provisioned by ShopVista.
- Business users will provide reporting requirements and validation feedback.

7. Timeline & Effort

Estimated project duration: 6–8 weeks.

Phases include setup, ingestion, transformation, validation, and dashboard delivery.